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Lab 10.2 Machine Learning 3 - Real-time Sound Recognition for Classification

CNN model deployment to Raspberry Pi

2.1 TensorFlow Lite

Based on the sample code in the previous section, let's apply the CNN model to predict the operational state of the vacuum pump. To do this, we will use TensorFlow Lite. TensorFlow Lite has limited features but is faster and lighter than TensorFlow. When you install TensorFlow on Raspberry Pi and laptop, TensorFlow Lite module is automatically installed. Because it supports Keras interface, TensorFlow Lite is able to load and run the CNN model developed in the prelab using TensorFlow and Keras.

The best way to practice implementation of the trained model is to program yourself. Because of the limited time in the lab, however, the complete sample code (lab10_sample2.py) is given. The simplified flowchart of the sample code is shown in Figure 5. Perform Task 2.1 to interpret the CNN model result (7th block - Algorithm to interpret CNN model result in Figure 5) as the appropriate label. Note that the CNN model (h5 format) must be in the same directory in which the sample code is.

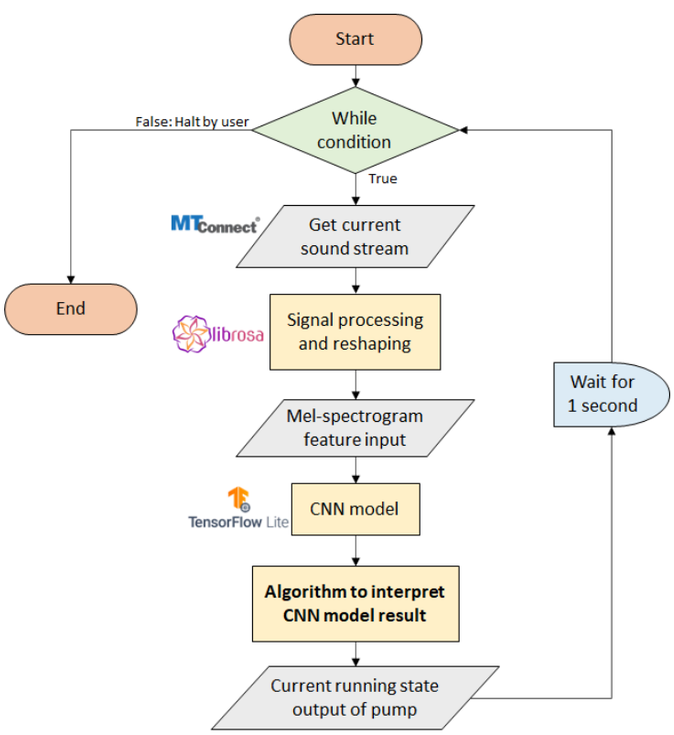

Figure 5 Flowchart of lab10_sample2.py

Task 2.1

Run lab10_sample2.py on your Raspberry Pi or laptop see if the CNN model works well. The terminal or output shell looks like as Figure 6. **Note that if you develop the CNN model with different feature size and shape, you have to modify the sample code.** An algorithm to interpret the CNN model output is incomplete as part of the sample code below. Complete the algorithm so that appropriate label is printed out at the end. To be specific, if Y (prediction inference index) is 0, the prediction label must be "OFF" (given), if Y is 1, the prediction label must be "ON-airleaking", and so on.

- Hint: Use if, elif, else statement.
- Hint: Another way is to use dictionary data structure with indexing.

Python - Python3 (Last part of lab10_sample2.py)

```
91        if Y == 0: # if Y is 0, do below
92            prediction_label = "OFF"
93        # complete your algorithm to take appropriate prediction_label string according to the CNN mo
94
95        print('The pump is now {}.\n'.format(prediction_label))
96
97        time.sleep(1) # pause for 1 second
```

```
Last Sequence: 65380
Timestamp: 2026-04-05T20:39:08.899582Z
Sound level: 44.91327120027029 dB
Model output: [[9.9826944e-01 1.6669695e-05 1.7138710e-03]]
Prediction inference index: 0
The pump is now OFF.

Last Sequence: 65406
Timestamp: 2026-04-05T20:39:10.018832Z
Sound level: 44.91273760834345 dB
Model output: [[9.9826956e-01 1.6669126e-05 1.7137879e-03]]
Prediction inference index: 0
The pump is now OFF.

Last Sequence: 65431
Timestamp: 2026-04-05T20:39:11.081565Z
Sound level: 45.24041555781059 dB
Model output: [[9.9826956e-01 1.6668808e-05 1.7138091e-03]]
Prediction inference index: 0
The pump is now OFF.

Last Sequence: 65457
Timestamp: 2026-04-05T20:39:12.195184Z
Sound level: 44.717398191167845 dB
Model output: [[9.9826968e-01 1.6668095e-05 1.7136770e-03]]
Prediction inference index: 0
The pump is now OFF.
```

Figure 6 Terminal while running lab10_sample2.py

```
## copy and paste your entire algorithm from line number 91 (if Y == 0: # if Y is 0, do below) to the
 if Y == 0:
            prediction_label = "OFF"

 elif Y == 1:
            prediction_label = "ON-airleaking"

 elif Y == 2:
           prediction_label = "ON-vacuum"

 else:
      prediction_label = "UNKNOWN"

        # complete your algorithm to take appropriate prediction_label string according to the CNN mod
        print(f"Prediction: {prediction_label} | Confidence: {np.max(yhat):.4f}")
        print('The pump is now {}.\n'.format(prediction_label))
        time.sleep(1) # pause for 1 second
```

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```
(base) pi@AaryaFarheen:~/prelab10 $ python3 lab10_sample2.py
Last Sequence: 61612
Timestamp: 2026-04-08T18:02:33.614501Z
Sound level: 43.8493692076777 dB
Model output: [[9.9980146e-01 3.6644925e-09 1.9851631e-04]]
Prediction inference index: 0
Prediction: OFF | Confidence: 0.9998
The pump is now OFF.

Last Sequence: 61749
Timestamp: 2026-04-08T18:02:39.600175Z
Sound level: 69.95127587832954 dB
Model output: [[4.0160157e-04 8.2571683e-03 9.9134129e-01]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9913
The pump is now ON-vacuum.

Last Sequence: 61780
Timestamp: 2026-04-08T18:02:40.985961Z
Sound level: 70.3158136430385 dB
Model output: [[1.8154969e-04 8.8572279e-03 9.9096125e-01]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9910
The pump is now ON-vacuum.

Last Sequence: 61813
Timestamp: 2026-04-08T18:02:42.461803Z
Sound level: 70.02979083040317 dB
Model output: [[0.00146597 0.00416505 0.994369  ]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9944
The pump is now ON-vacuum.

Last Sequence: 61846
Timestamp: 2026-04-08T18:02:43.935500Z
Sound level: 70.22699214195691 dB
Model output: [[1.2066163e-04 1.0326604e-02 9.8955274e-01]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9896
The pump is now ON-vacuum.

Last Sequence: 61879
Timestamp: 2026-04-08T18:02:45.394709Z
Sound level: 70.35739541432177 dB
Model output: [[1.6791878e-04 7.2130221e-03 9.9261910e-01]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9926
The pump is now ON-vacuum.

Last Sequence: 61914
Timestamp: 2026-04-08T18:02:46.941795Z
Sound level: 70.2133037566 dB
Model output: [[4.7264918e-04 4.9506081e-03 9.9457669e-01]]
Prediction inference index: 2
Prediction: ON-vacuum | Confidence: 0.9946
The pump is now ON-vacuum.
```